

The Islamia University of Bahawalpur
Department of Information Technology



SOFTWARE REQUIREMENTS SPECIFICATION
(SRS DOCUMENT)

for

E-Commerce Store Web

By

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Revision History

Name	Date	Reason for changes	Version
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Application Evaluation History

Comments (by committee) *include the ones given at scope time both in doc and presentation	Action Taken
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Supervised by
< Mr. Muzammil Ul Rahman >

Signature

1.1 Introduction

The E-Commerce Store Website project is designed to create a dynamic and user friendly online platform for buying and selling a wide range of products. This project focuses on developing a seamless shopping experience for users, while providing robust tools for merchants to manage their inventory and sales. The website will feature an intuitive user interface (UI) and user experience (UX), ensuring easy navigation and accessibility across various devices.

1.1.1 Purpose

The objective of this E-Commerce Store Web project is to create an interactive and easy-to-use platform that facilitates online purchases and sales. This system will:

1. Let customers seamlessly browse, search, and buy products.
- Provide the merchant with necessary tools that enable effective inventory management, tracking of orders, and sales monitoring.
2. Enable the administrator to have a look at the entire system with regard to smooth operation and data security.
3. The platform aims to close the gap between customers and businesses, providing an intuitive shopping experience while utilizing modern web technologies for performance, scalability, and security.

1.1.2 Scope

The E-Commerce Store Website is aimed to be a user-friendly shopping platform. It offers essential features like product browsing, cart management, secure checkout, and user account management. The platform emphasizes smooth customer shopping and efficient

merchant inventory management and sales tracking.

Key Features:

User Interface: Easy navigation for product discovery and a robust product management system for merchants.

User Accounts: Secure signup, login, and customized profile management.

Admin Dashboard: Tools for order management, sales analytics, and customer support.

Performance Optimization: Caching and using content delivery networks (CDNs) to ensure super-fast page loading.

Security: Using SSL encryption and proper authentication mechanisms to ensure the safety of user data.

Excluded: The project does not include any advanced customizations, enterprise system integration, or mobile application development. The project will be a fully responsive web design.

1.2 Overall description

1.2.1 Product perspective

The E-Commerce Web Store is an entirely new product that would provide a comfortable and safe online shopping. It will be designed both for the customers and for the merchants, with options like a seamless browsing experience, secure checkout, and total product management. The entire system will be built with scratch as per the requirement of the modern demands for e-commerce to be scalable, secure, and efficient.

1.2.2 Operating environment

The E-Commerce Web Store will be implemented fully in a JavaScript environment based on the following technologies:

Frontend: ReactJS is used to build the application's user interface, thus giving it a dynamic and responsive feel on all devices: mobile, tablet, desktop.

Backend: NodeJS. This is the runtime environment for executing JavaScript server-side, handling requests, and managing server operations.

ExpressJS. This is a web framework built on top of NodeJS that streamlines development of RESTful APIs meant for handling client-server interaction.

Database:

MongoDB: It is a NoSQL database to store and manage the product, order, and user data. MongoDB has been chosen for its scalability and flexibility in handling large data volumes.

Hosting & Deployment:

Servers: The application will be hosted on cloud platforms such as AWS or Heroku, providing reliable uptime and scalability.

Geographic Locations: Servers and databases can be hosted in any geographic location in order to minimize latency across the globe.

Web Browsers: The system should function well on the latest versions of Google Chrome, Mozilla Firefox, Safari, and Microsoft Edge.

Compatibility with older browsers like Internet Explorer will be limited or unsupported based on modern web standards.

Operating Systems: Windows, Linux, and macOS will support development and deployment. This platform will be accessed with modern browsers for cross-browser compatibility.

1.2.3 Design and implementation constraints

Programming Language:

The system must be developed using JavaScript for full-stack development (ReactJS, NodeJS, ExpressJS), ensuring smooth front-end and back-end communication.

Database:

The platform will use MongoDB due to its flexibility, scalability, and ability to handle unstructured data effectively.

Hosting Platform:

The application will be hosted on AWS or Heroku to ensure scalability and reliable performance.

Browser Compatibility:

The platform must support modern browsers such as Chrome, Firefox, Safari, and Edge, ensuring compatibility with modern web standards.

Security:

SSL encryption and OAuth 2.0 will be implemented to ensure secure transactions and user data protection.

1.3 Requirement identifying technique

This section describes the requirements identifying technique(s) which further help to derive functional requirements specification. This is focus on building interactive library management system. For instance,

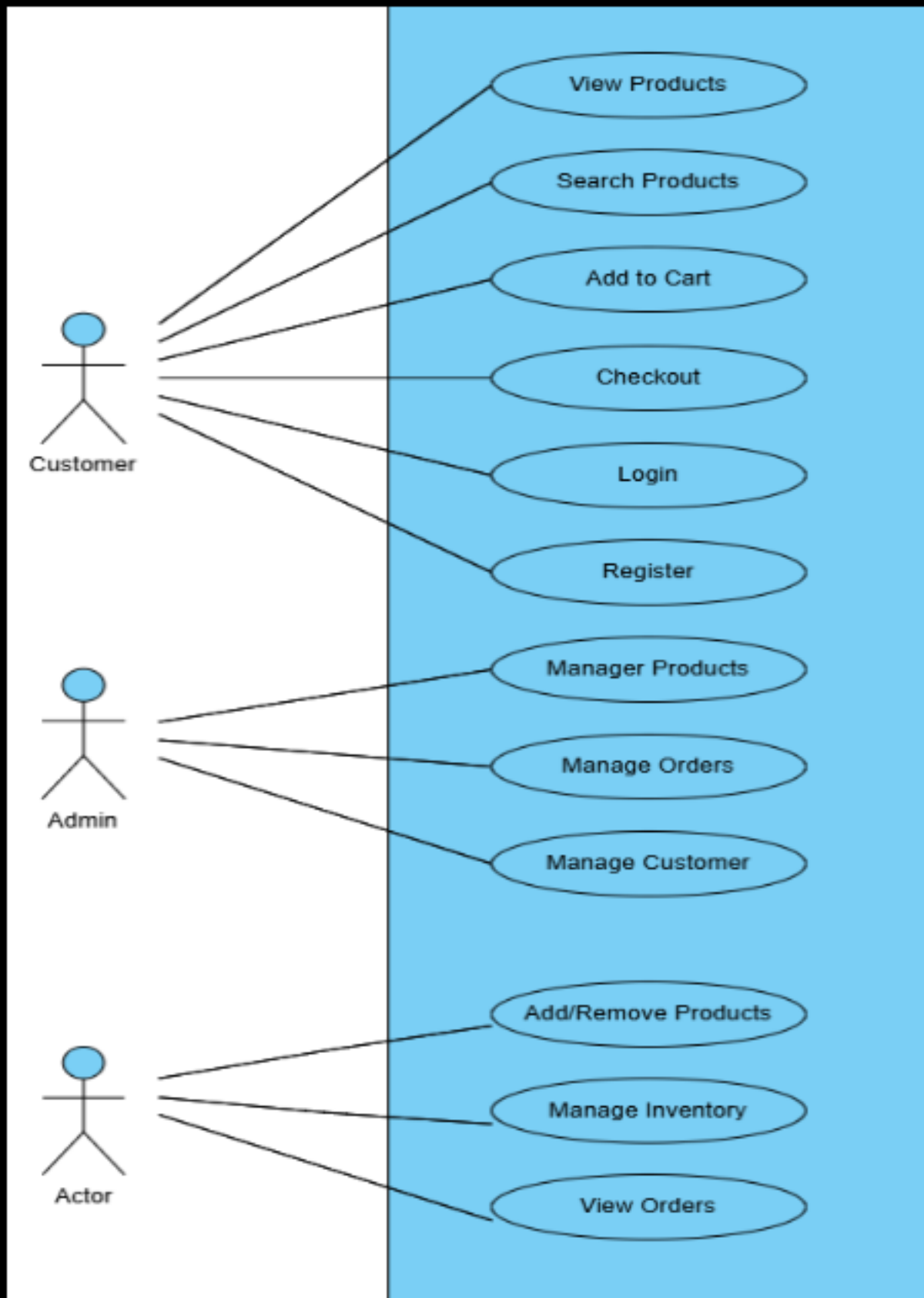
Use case is technique to identify interactions between users and the system.

Event- response tables is for real time system include book requests event posting, book searches.

Story boarding for graphically intensive applications and user friendly interface.

In addition to the above, projects involving data warehouses, batch processes, hardware devices with embedded control software, and computationally intensive applications require specialized techniques for requirement identification. These techniques are more suited for complex systems involving large-scale data handling or hardware integration and are beyond the scope of this project.

1.3.1 Use case diagram



1.3.2 Use case description

This use case enables customers to order products from the online store, check for availability, and complete the checkout process. After purchase, the status of the product is marked as "sold out," and if the item is returned, its status is marked again as "available" for other customers. The system also handles out-of-stock items and provides notifications for restocks. The Admin or Super Admin handles the product inventory, keeping the stock levels accurate, and cleans out outdated or discontinued products from the catalog.

Table 5 Show the detail use case template

Use Case ID	JC-1
Use Case Name	Register/Login
Actors	Customer
Description	A customer registers a new account or logs into an existing account.
Trigger	The customer accesses the website and selects Login/Register
Preconditions	PRE-1: Customer has an email or social media account for login.

Postconditions POST-1: Customer is successfully logged in and redirected to the homepage.

Normal Flow

1. Customer clicks on Login/Register.
2. System prompts for credentials or registration details.
3. Customer enters the information.
4. System verifies credentials or creates a new account.
5. Customer is logged in or registered.

Alternative Flow

- 1.1 Incorrect Credentials: Customer enters wrong credentials, and the system displays an error message.
- 1.2 Password Reset: Customer chooses to reset the password.

Exceptions

E1: System fails to verify credentials due to network issues.

Business Rules

BR-1: Valid credentials must be used for login.

Assumptions

A1: Assume most customers log in with email.

Use Case ID UC-2

Use Case Name	Browse Products
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Actors	Customer
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Description	The customer browses the available product catalog.
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Trigger	Customer accesses the homepage or product catalog.
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Preconditions	PRE-1: Products exist in the catalog.
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Postconditions	POST-1: Customer views product details or selects products.
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Normal Flow

1. Customer accesses the product catalog.
2. System displays products with search and filter options.
3. Customer browses or searches products.
4. System displays product details upon selection.

Alternative Flow

- 1.1 Out-of-Stock Products: Products display "Out of Stock" status.

Exceptions

- E1: System fails to load the product catalog due to an error.

Business Rules

- 3R-1: Products must have valid availability statuses.

Assumptions

- A1: Assume customers prefer searching products using filters.

Use Case ID

UC-3

Use Case

Name Add Item to Cart

Actors

Customer

Description

Customer selects a product and adds it to their shopping cart.

Trigger Customer clicks "Add to Cart" for a product.

Preconditions PRE-1: Product is available in stock.

Postconditions

POST-1: Product is added to the shopping cart.

Normal Flow

1. Customer views a product.
2. Customer clicks "Add to Cart".
3. System adds the product to the cart.
4. System updates cart details.

Exceptions E1: Product fails to add due to technical errors.

Business Rules

BR-1: Cart must allow only available products to be added.

Assumption

A1: Assume customers add 1-5 items per session.

Use Case

ID

JC-4

Use Case

Checkout and Make Payment

Name

Actors

Customer, Payment Gateway

Description

Customer completes the purchase process and makes payment.

Trigger

Customer proceeds to checkout from the cart.

Preconditions

PRE-1: Customer has valid items in the cart.

PRE-2: Payment method is valid.

Postconditions

POST-1: Order is confirmed.

POST-2: Payment is processed.

Normal Flow	1. Customer proceeds to checkout. 2. System displays order details. 3. Customer selects a payment method. 4. Payment gateway processes the payment. 5. System confirms the order and updates inventory.
Alternative Flow	1.1 Payment Failure: Customer retries or chooses a new payment method.
Exceptions	E1: Payment gateway downtime prevents payment processing.
Business Rules	3R-2: Payment transactions must be secure.
Assumptions	A1: Assume 40% of customers choose expedited shipping.
Use Case ID	UC-5

Use Case Name	View Order History
Actors	Customer
Description	Customer views their past orders and order statuses.
Trigger	Customer selects "Order History" from their profile.
Preconditions	PRE-1: Customer has placed at least one order.
Postcondition	POST-1: Customer views their order details.
Normal Flow	1. Customer accesses "Order History". 2. System displays past orders. 3. Customer selects an order to view its status and details.
Alternative Flow	1.1 No Orders: System informs the customer that no orders exist.
Exceptions	E1: System fails to fetch order history due to an error.
Business Rules	BR-1: Order details must include order date, items, and status.
Assumptions	A1: Assume customers frequently check recent orders.

Use Case

ID **JC-6**

Use Case

Name **Manage Products**

Actors

Admin

Description Admin adds, updates, or removes products from the system.

Trigger

Admin accesses the "Manage Products" section.

Preconditions

PRE-1: Admin is logged into the system.

Postconditions	POST-1: Product catalog is updated.
Normal Flow	1. Admin selects a product. 2. Admin performs add, update, or delete action. 3. System updates the product catalog.
Alternative Flow	1.1 Invalid Details: System rejects invalid product data.
Exceptions	E1: System fails to update due to technical issues.
Assumptions	A1: Assume admins update products weekly.

Use Case ID	UC-7
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Use Case Name	Manage Users
Actors	Admin

Description Admin manages user accounts (edit, delete).

Trigger Admin accesses "Manage Users" section.

Preconditions
PRE-1: Admin is logged in.

Postconditions
POST-1: User accounts are updated.

Normal Flow
1. Admin selects a user.
2. Admin edits or removes user details.
3. System confirms updates.

Exceptions
E1: System fails to update user records.

Use Case ID UC-8

Use Case Name Approve Orders and Update Inventory
Actors Merchant, Inventory System

Description Merchant approves orders and updates product inventory.

Trigger

Order is placed or updated.

Preconditions

PRE-1: Order exists in the system.

Postconditions

POST-1: Inventory is updated.

POST-2: Order status is approved.

1.4 Functional Requirements

User Account Management

Customers can register, log in, and update their profile.

Passwords are securely stored with encryption.

Users can reset forgotten passwords via email.

Product Catalog

Customers can view products with detailed information and filter by price, category, and rating.

Admins can manage the product catalog, including adding, editing, or removing items.

Product availability is displayed for customers.

Shopping Cart

Customers can add, remove, and edit items in their cart.

The cart calculates the total price, including taxes and shipping.

Cart contents persist across sessions for logged-in users.

Checkout and Payment

Customers can proceed with checkout, select shipping and payment methods.
Payments are securely processed via third-party gateways.
The system generates an order confirmation and sends an email notification to the customer.

Order Management

Customers can view order history and status.
Admins can manage, modify, and process returns for orders.
Admins can update order statuses (shipped, delivered, canceled).

Inventory Management

Product availability is automatically updated after an order is placed.
Admins are notified when stock levels fall below a threshold.
Admins can add, update, and remove inventory items.
Backorders are handled by allowing customers to place orders for out-of-stock items.

Admin Panel

Admins can manage users, generate reports, and configure website settings.
Role-based access control is provided for different system functionalities.

Security

Sensitive data is transmitted securely using encryption (SSL/TLS).
Multi-factor authentication (MFA) is implemented for Admin access.
User activity, especially Admin actions, is logged for security auditing.

Notification System

Email notifications are sent for order confirmations, shipping updates, and product availability.
Alerts are sent to Admins for inventory updates or low stock warnings.

Customer Support

A support portal is available for customers to submit inquiries or complaints.

Admins can manage support tickets and integrate with third-party tools for real-time support.

Table 6 Show the functional requirement template

Identifier	1.1
Title	User Registration
Requirement	The customer shall be able to register with their email address and password.
Source	Business Analyst
Rationale	To allow new users to create an account and access features like shopping carts and order history.
Business Rule(if required)	Password must be encrypted.
Dependencies	None
Priority	High
Identifier	1.2
Title	Password Recovery
Requirement	The system shall allow users to reset forgotten passwords through email verification.
Source	Customer Feedback
Rationale	To ensure users can recover account access when they forget their credentials.
Business Rule(if required)	Email must be valid and verified.
Dependencies	None
Priority	High
Identifier	2.1
Title	Browse Products

Requirement	Customers shall be able to view and search for products based on filters like price, category, and ratings.
Source	Product Manager
Rationale	To enable customers to easily find products of interest.
Business Rule(if required)	Nope
Dependencies	Nope
Priority	High

Identifier	3.1
Title	Shopping Cart
Requirement	The customer shall be able to add, remove, or update quantities of items in their shopping cart.
Source	Customer Feedback
Rationale	To provide a flexible shopping experience.
Business Rule(if required)	None

Dependencies	2.1
Priority	High

Identifier	4.1
Title	Checkout
Requirement	The system shall allow customers to proceed to checkout and select a shipping and payment method.
Source	Business Analyst
Rationale	To complete the order process seamlessly and securely

Business Rule(if required)	Payment must be processed securely.
Dependencies	3.1
Priority	High
Identifier	5.1
Title	Inventory Update
Requirement	The system shall automatically update product availability when a customer places an order.
Source	Business Analyst

Rationale	To ensure accurate inventory tracking.
Business Rule(if required)	Nope
Dependencies	4.1
Priority	High
Identifier	7.1
Title	Admin Role Management
Requirement	The system shall restrict Admin functionalities based on assigned roles.
Source	Admin Feedback
Rationale	To prevent unauthorized actions by Admin users.
Business Rule(if required)	Roles must be clearly defined.
Dependencies	Nope
Priority	High

Identifier	3.1
Title	Secure Data Transmission
Requirement	The system shall transmit sensitive data, such as passwords and payment details, securely using SSL/TLS encryption.
Source	Security Specialist
Rationale	To protect user data and prevent data breaches.
Business Rule(if required)	SSL/TLS must be implemented
Dependencies	Nope
Priority	High

Identifier	10.1
Title	Support Ticket System
Requirement	Customers shall be able to submit inquiries or complaints through a support portal.
Source	Customer Feedback
Rationale	To provide an efficient way for customers to communicate their issues or questions.
Business Rule(if required)	Tickets must be acknowledged in 24 hrs.
Dependencies	Nope
Priority	Medium

1.5 Non-Functional Requirements

Non-functional requirements (NFRs) describe the quality attributes of a system, such as performance, security, reliability, scalability, usability, and compliance. These requirements specify how the system should behave rather than what it should do. NFRs are crucial for ensuring that the system meets the desired quality standards and can handle real-world demands. Below is a list of example non-functional requirements that are applicable to an e-commerce web store:

Example Non-Functional Requirements for an E-Commerce Web Store:

Performance

- NFR-1.1: The system shall support 1,000 concurrent users with no performance degradation.
- NFR-1.2: The system shall load any product page within 2 seconds under normal load conditions.
- NFR-1.3: The checkout process shall complete within 5 seconds, including payment processing, under typical conditions.

Scalability

- NFR-2.1: The system shall be able to handle up to 10,000 concurrent users without any significant performance issues.
- NFR-2.2: The system shall scale horizontally to handle increased traffic during peak times (e.g., holiday sales).

Availability

- NFR-3.1: The system shall maintain 99.9% uptime annually, allowing for no more than 8.76 hours of downtime per year.
- NFR-3.2: Maintenance windows shall be scheduled in advance and shall not exceed 1 hour during non-peak hours.

Security

- NFR-4.1: The system shall encrypt all sensitive data (e.g., passwords, payment details) using SSL/TLS encryption.
- NFR-4.2: The system shall implement multi-factor authentication (MFA) for admin and super admin logins.
- NFR-4.3: All credit card transactions shall comply with PCI DSS standards.

Usability

- NFR-5.1: The website shall have a responsive design, ensuring that it works on devices ranging from desktop computers to smartphones.
- NFR-5.2: The checkout process shall require no more than 4 steps for users to complete their purchase.
- NFR-5.3: Customers should be able to find products via search within 5 seconds.

Reliability

- NFR-6.1: The system shall recover from failures within 5 minutes with no data loss.
- NFR-6.2: The system shall ensure 100% accuracy in order processing and inventory management.

Maintainability

- NFR-7.1: The system shall provide a modular architecture that allows for easy updates and additions of new features.
- NFR-7.2: The system's codebase shall follow best practices for documentation, enabling ease of understanding for future developers.

Compliance

NFR-8.1: The system shall comply with GDPR regulations for data protection and user privacy.

NFR-8.2: The system shall provide mechanisms for users to request deletion of their personal data in accordance with privacy laws.

Localization

NFR-9.1: The system shall support multiple languages (e.g., English, Spanish, French) for global users.

NFR-9.2: The system shall allow for currency conversion based on the user's location.

Backup and Recovery

NFR-10.1: The system shall perform full backups every 24 hours, with incremental backups every hour.

NFR-10.2: The system shall allow for recovery of data up to 30 days from any backup.

1.5.1 Summary of Usability Requirements

The system usability requirements ensure an effective user experience by the proper consideration of the following:

Ease of Learning : The system should be straightforward for new users to grasp and use.

Ease of Use: The user interface should be intuitive. As little effort as possible must be made to execute operations.

Example: A user should be able to retrieve their previous meal orders with a single click. (e.g., USE-1)

Error Avoidance: The system should help users avoid making errors by providing clear instructions and warnings.

Example: Confirmation prompts before submitting an order to prevent mistakes.

Error Recovery: The system should enable users to easily fix errors or undo actions without major consequences.

Example: Users should be able to edit their cart items before finalizing the order.

Efficiency of Interactions: The system should enable users to accomplish tasks in the least number of steps.

Example: The checkout process should be streamlined so that the user has to enter minimal

information.

Accessibility: The system should be usable by people with diverse abilities, including those with disabilities.

Example: The system should support screen readers and allow navigation using keyboard shortcuts.

1.5.2 Performance Requirements Summary:

Performance requirements specify how the system should behave in terms of speed, responsiveness, and efficiency during various operations. These requirements ensure that the system can handle expected loads, respond quickly, and perform tasks within acceptable timeframes.

Example performance requirement:

PER-1: 95% of webpages generated by the COS shall download completely within 4 seconds from the time the user requests the page, assuming a 20 Mbps or faster internet connection.

These performance goals should be integrated with the corresponding functional requirements for clarity and alignment.

1.6 References

References may include any documents, standards, or guidelines that inform the system design or performance expectations

"A Review Paper on E-commerce"

This paper provides an overview of the current scenario of e-commerce, analyzing trends and developments in the field. [ResearchGate](#)

"Exploring the Benefits of E-commerce Applications for Efficient Online Operations"

This research explores how e-commerce applications can enhance operational efficiency, discussing tools for managing inventory, shipping, and taxes. [ResearchGate](#)

"Research on E-commerce Platform of Online Shopping Consumers"

This study analyzes consumer purchasing tendencies across various e-commerce platforms, providing insights into consumer behavior.

[Atlantis Press](#)

